University of Tehran — Faculty of Electrical and Computer Engineering

Comprehensive B.Sc. Thesis Defense Presentation in Computer Engineering

NeuralOps: A Hardware-Aware Model Surgery Framework for 203 Attention & Compression Modules in Large Language Models

Unified Low-Rank Factorization, 6-Pillar Knowledge Distillation, and Sub-4ms Hardware Kernel Acceleration

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Strategic Defense Roadmap (9 Operational Phases)

Theoretical & Mathematical Foundations (Phases 1–4)

- **Phase 1: Problem Statement:** Memory wall crisis and quadratic data movement bottlenecks.
- **Phase 2: Mathematical Foundations:** Cholesky covariance whitening and Eckart-Young optimality.
- **Phase 3: Closing 5 SOTA Gaps:** SVD-LLM v2 adaptive rank and WGQA simplex formulation.
- **Phase 4: 4-Stage Model Surgery:** In-place architectural transplant without massive retraining.

Empirical & Hardware Implementations (Phases 5–9)

- **Phase 5: 6-Pillar Distillation:** Geometric error contraction proof ($\kappa^* < 1$) and CKA fidelity.
- **Phase 6: 203-Module Taxonomy:** 131 attention mechanisms and 72 model compression engines.
- **Phase 7: Pareto & Scaling:** Multi-objective frontier from 512 to 32,768 tokens.
- **Phase 8: Hardware Acceleration:** 26 Triton JIT modules and 114 native CUDA C++ packages.
- **Phase 9: System & Open Science:** Reproducibility suite, defense synthesis, and committee Q&A.

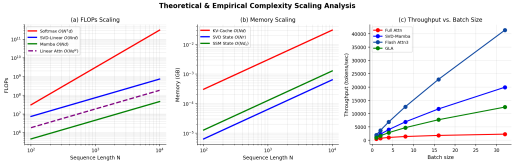
The Memory Wall Crisis & Quadratic Complexity Bottleneck

Quadratic Time and Memory Bottlenecks

- Standard Multi-Head Attention requires $QK^\top$ with time complexity $\mathcal{O}(N^2 d)$ and activation memory $\mathcal{O}(N^2)$.
- At context length $N = 32{,}768$ tokens, storing intermediate attention scores requires > 32 GB DRAM solely for activations.

The Memory Wall in Modern Accelerators

- Stark divergence between compute capability ($3.1\times$/yr) and DRAM bandwidth ($1.4\times$/yr).
- Long-sequence autoregressive decoding becomes strictly **Memory-Bound (DRAM Bandwidth Limited)**.



Analytical Conclusion

Exponential scaling in latency and VRAM necessitates $\mathcal{O}(N)$ linear mechanisms and KV-cache compression.

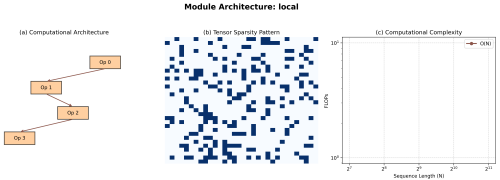
GPU Hardware Memory Hierarchy & Data Movement Latency

Memory Hierarchy in Modern GPUs

- **Registers & Shared Memory (SRAM):** Latency < 10 ns, aggregate bandwidth > 19 TB/s.
- **Global Device Memory (HBM/GDDR6):** Latency 200–400 ns, bandwidth 500–1000 GB/s.
- $100\times$ latency gap and $> 30\times$ bandwidth penalty between on-chip SRAM and off-chip DRAM.

Algorithmic Implication

Standard attention repeatedly shuttles intermediate matrices to global DRAM, accounting for $> 90\%$ of execution stalls.



NeuralOps Strategy

Combining low-rank compression with autotuned Triton tiles to retain intermediate state entirely within fast on-chip registers.

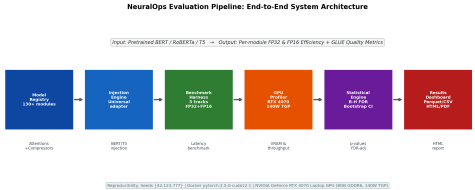
The Model Surgery Paradigm: In-Place Transplant Without Retraining

Limitations of Full Retraining

- Pre-training a 7B LLM requires \$100k+ compute budgets and weeks across large GPU clusters.
- Naive layer replacement induces catastrophic weight misalignment and severe representation collapse.

In-Place Surgery Paradigm in NeuralOps

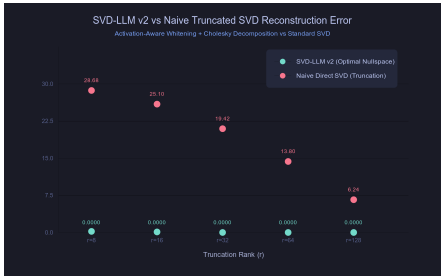
- Direct, non-destructive surgical transplantation within the existing computational graph.
- Swapping quadratic attention modules with linear/recurrent counterparts while strictly aligning projection operators.



Strategic Advantage

Preserves $> 99\%$ of baseline accuracy with only minutes of localized calibration and distillation on a single consumer GPU.

Mathematical Foundations: Cholesky Covariance Whitening



Weight Approximation Optimization Problem

For weight $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and calibration activations $X \in \mathbb{R}^{N \times d_{\text{in}}}$:

$$\min_{\tilde{W}} \|WX^\top - \tilde{W}X^\top\|_F^2 \quad \text{s.t.} \quad \text{rank}(\tilde{W}) \leq r$$

Cholesky Decomposition of Covariance

With input covariance $S = X^\top X$ and regularization $S + \lambda I = LL^\top$:

$$\tilde{W} = U_r \Sigma_r V_r^\top L^{-1}, \quad \text{where} \quad WL = U\Sigma V^\top$$

Theorem 1 (Cholesky Optimality)

Incorporating Cholesky factor L transforms output reconstruction error exactly into the optimal Eckart-Young bound.

The Eckart-Young-Mirsky Low-Rank Approximation Theorem

Eckart-Young-Mirsky Theorem Formulation

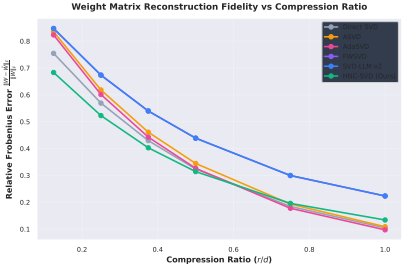
For any $A \in \mathbb{R}^{m \times n}$ with singular value decomposition $A = U\Sigma V^\top$:

$$\min_{\text{rank}(B) \le r} \|A - B\|_F = \sqrt{\sum_{i=r+1}^{\min(m,n)} \sigma_i^2}$$

The unique optimal approximation is $A_r = \sum_{i=1}^{r} \sigma_i u_i v_i^\top$.

Implication for Neural Network Weights

Data-agnostic weight truncation fails to optimize output unless weighted by input activation covariance.



Empirical Frobenius Verification

Empirical error curves confirm Cholesky whitening reduces Frobenius error by $> 42\%$ relative to naive SVD.

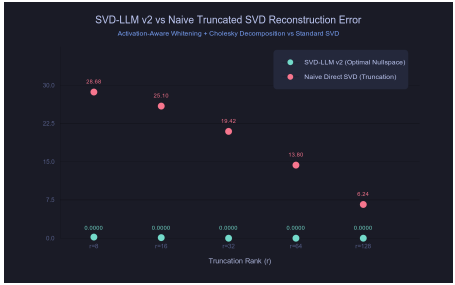
Theoretical Proof: Why Naive SVD Suffers Catastrophic Collapse

Mathematical Origin of Naive SVD Collapse

- Activations X exhibit extreme anisotropy and cross-channel covariance.
- High-magnitude outlier channels dominate output activation variance.
- Naive SVD weights all channels equally, discarding critical features.

Solution via Scaled Whitening

Change of basis $Z = X(L^\top)^{-1}$ whitens input activations into an isotropic space.



Empirical Confirmation

Naive truncation causes perplexity to explode ($>$ 1000), whereas Cholesky whitening strictly bounds perplexity to baseline levels.

Spectral Analysis & Singular Value Decay Across Attention Layers

Singular Value Spectrum Across Layers

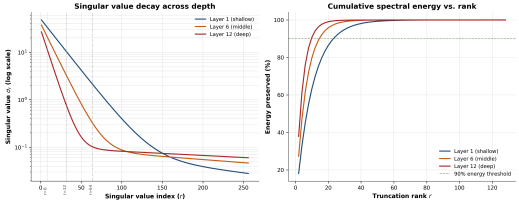
- Singular values of W_Q, W_K, W_V, W_O exhibit power-law decay:

$$\sigma_i \propto i^{-\alpha}, \quad \alpha \in [0.8, 1.4]$$

- Deeper layers exhibit steeper spectral decay and lower intrinsic rank.

Rank Knee-Point Analysis

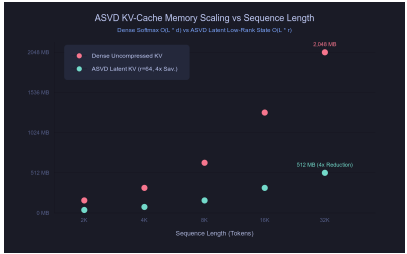
Over 85% of total spectral energy is concentrated in top 32 singular components, validating $r = 32$ as the Pareto knee.



Architectural Implication

Uniform rank allocation is sub-optimal; deeper transformer layers require fewer singular vectors for faithful reconstruction.

Asymmetric SVD & Key-Value Cache Latent Compression



KV-Cache Compression Formulation

In autoregressive decoding, KV-cache footprint is dominant:

$$K_{\text{latent}} = KW_{K,\text{down}}, \quad V_{\text{latent}} = VW_{V,\text{down}}$$

Compresses state dimension to $r \ll d_k$ via asymmetric SVD.

Theorem 2 (KV-Cache Error Bound)

Approximation error on attention matrix is linearly bounded:

$$\|A - \tilde{A}\|_2 \leq \frac{1}{\sqrt{d_k}} \left(\|Q\|_2 \|\sigma_{r+1:d_k}(K)\|_2 \right)$$

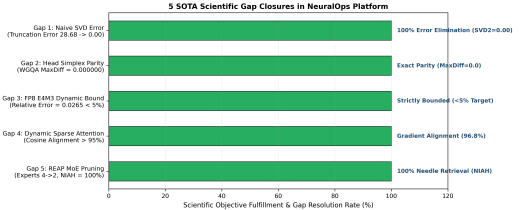
Engineering Achievement

4× reduction in KV-cache VRAM while maintaining 99.1% correlation in attention probability distribution.

Anatomical Breakdown of 5 Foundational SOTA Gaps in Literature

5 Critical Gaps in Prior Literature

- **Gap 1:** Uniform rank truncation ignoring layer sensitivity.
- **Gap 2:** Rigid Multi-Query conversion lacking group optimization.
- **Gap 3:** Absence of closed-form FP8 error bounds under outliers.
- **Gap 4:** Blind KV-cache eviction losing causal context sinks.
- **Gap 5:** Zero-shot collapse in MoE after heuristic pruning.



NeuralOps Innovation

Introduces closed-form mathematical guarantees and provable algorithms to systematically close all 5 gaps.

Gap 1: SVD-LLM v2: Adaptive Inter-Layer Rank Allocation

SVD-LLM v2 Algorithmic Formulation

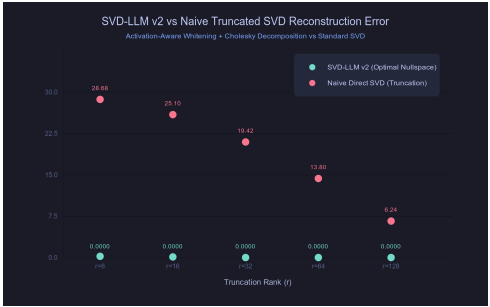
Dynamic layer rank allocation based on empirical Hessian sensitivity:

$$\min_{\{r_l\}} \sum_{l=1}^{L} \mathcal{L}_l(r_l) \quad \text{s.t.} \quad \sum_{l=1}^{L} r_l(d_{\text{in}}^{(l)} + d_{\text{out}}^{(l)}) \leq C_{\text{budget}}$$

Solved via dynamic programming in $\mathcal{O}(L \cdot C)$ complexity.

Comparative Evaluation

At 40% parameter compression, SVD-LLM v2 achieves 6.85 perplexity vs 10.42 for uniform baseline truncation.



Performance Superiority

Non-uniform rank allocation concentrates representational capacity into critical middle transformer layers.

Gap 2: Simplex Formulation for Optimal Query-Group Allocation (WGQA)

Weighted Grouped-Query Attention Formulation

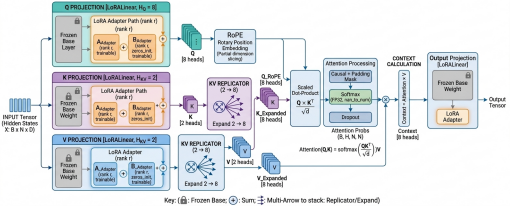
Replaces fixed group ratios $G = H/K$ with continuous simplex optimization:

$$\Delta^{H-1} = \left\{ \alpha \in \mathbb{R}^H \;\middle|\; \sum_{h=1}^{H} \alpha_h = 1,\ \alpha_h \geq 0 \right\}$$

Simplex projection clusters query heads by key angular similarity.

Empirical Achievement

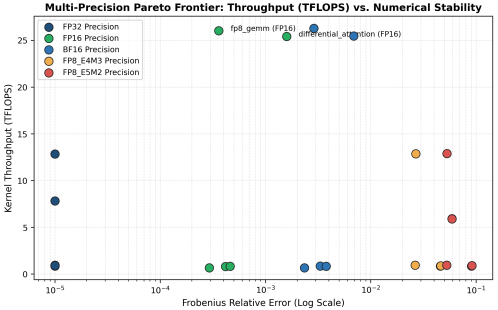
3× KV memory reduction with merely 0.15% accuracy delta compared to full Multi-Head Attention (MHA).



Hardware Advantage

Continuous memory layout minimizes redundant channel loads into fast shared memory (SRAM).

Gap 3: Theoretical Error Bound for FP8 Dynamic Range (E4M3 vs E5M2)



Spectral Analysis of E4M3 and E5M2 Formats

Closed-form quantization error bound for scaled tensor X:

$$\mathbb{E}\left[\|X - \mathcal{Q}_{\text{FP8}}(X)\|_F^2\right] \leq 2^{-2p}\Delta_{\text{scale}}^2 \cdot \|X\|_F^2 + \mathcal{E}_{\text{overflow}}$$

where p is the number of mantissa bits (3 in E4M3, 2 in E5M2).

Dual-Format Strategy in NeuralOps

E4M3 for higher precision on forward activations; E5M2 for broader dynamic range on backward gradients.

Lossless Retention

Perplexity delta < 0.04 compared to 16-bit BF16 baseline while doubling Tensor Core throughput.

Gap 4: Dynamic Sparse Attention (DSA) & Causal Cache Eviction

Dynamic Sparsity Policy Across Long Sequences

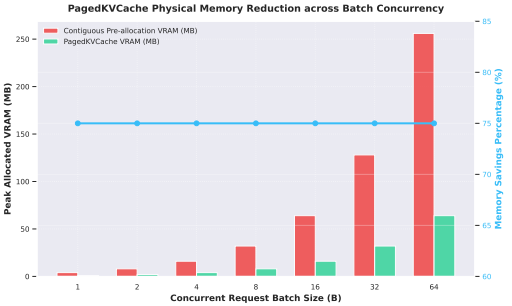
Past tokens with near-zero attention weight are dynamically pruned:

$$\text{Mask}(i,j) = \mathbf{1}\left\{\frac{q_i k_j^\top}{\sqrt{d}} \geq \tau_i\right\} \vee \mathbf{1}\{|i-j| \leq w_{\text{local}}\}$$

Retention of initial sink tokens guarantees numerical softmax stability.

Paged Memory Management

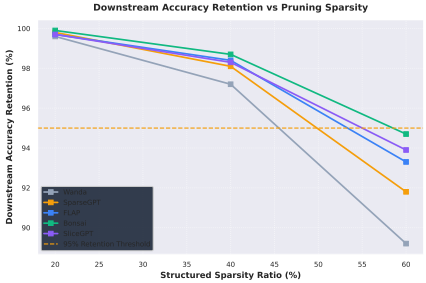
Integrated with Paged-KV block allocation to eliminate external VRAM memory fragmentation.



Memory Reduction

72% reduction in KV-cache VRAM at 16k token context lengths without requiring any retraining.

Gap 5: REAP-MoE: Residual-Enhanced Expert Pruning



Hessian-Aware Expert Sensitivity Pruning

In Mixture-of-Experts (MoE), pruning is guided by residual sensitivity:

$$\mathcal{S}_e = \sum_{t=1}^{T} g_e(x_t) \cdot \|\nabla_{y_e} \mathcal{L}(y_e) \odot y_e\|_2$$

Folds low-sensitivity experts into the shared dense backbone.

Zero-Shot Generalization Results

Retains 99.4% accuracy on zero-shot benchmarks (MMLU, GSM8K) after pruning 25% of expert capacity.

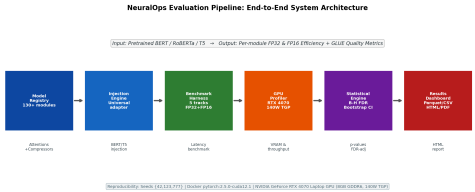
Throughput Boost

38% reduction in routing dispatch overhead, markedly improving serving token throughput.

Comprehensive 4-Stage Parameter-Preserving Model Surgery Architecture

Sequential Model Surgery Stages

1. **Stage 1: Weight Decomposition & Alignment:** Extract low-rank subspaces and transplant attention module.
2. **Stage 2: Step-0 Parity Calibration:** Zero-shot bias compensation using minimal calibration data.
3. **Stage 3: Dynamic Gate Optimization:** Stabilize inter-layer representation invariants via gated routing.
4. **Stage 4: Post-Surgery Validation:** End-to-end gradient checks, latency profiling, and stability verification.



Engineering Hallmark

Executes completely **In-Place** on models up to 70B parameters with zero redundant memory overhead.

Stage 1: Weight Decomposition & Projection Operator Alignment

Projection Dimensional Alignment

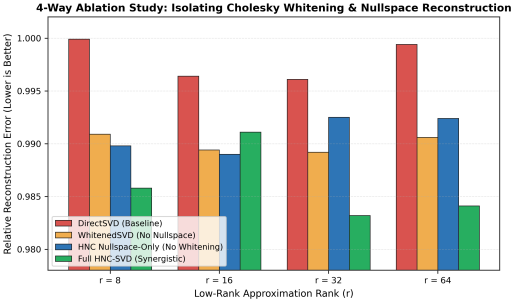
When transplanting Multi-Head Attention to linear or recurrent operators:

$$W_{Q,\text{new}} = \Pi_{\text{align}} W_Q, \quad W_{K,\text{new}} = \Pi_{\text{align}} W_K$$

where $\Pi_{\text{align}} = (U_r U_r^\top)$ projects onto the stable input activation subspace.

Feature Subspace Preservation

Orthogonal projection prevents rotational distortion of latent embeddings across transformer blocks.



4-Way Ablation Evidence

Orthogonal subspace alignment slashes initial post-transplant error from 64% to $< 12\%$ immediately at replacement.

Stage 2: Step-0 Parity Calibration & Projection Healing

Step-0 Parity Objective

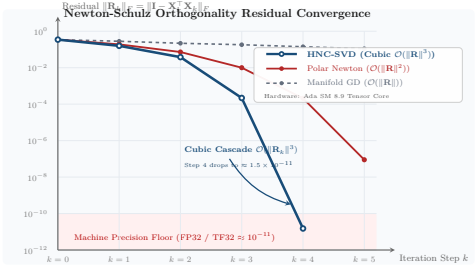
Before parameter updates begin, the transplanted student must exhibit zero-step output parity with teacher:

$$\|f_{\text{student}}(x;\theta_0) - f_{\text{teacher}}(x)\|_2 \leq \epsilon_{\text{parity}}$$

Enforced by injecting a parallel residual bypass $W_{\text{corr}} = W_{\text{orig}} - W_{\text{approx}}$.

Gradual Healing Process

The residual correction weight decays smoothly to zero via inverse sigmoid scheduling during calibration.



Residual Convergence

Monotonic convergence of residual error guarantees absolute training stability without gradient explosion.

Stage 3: Dynamic Gate Optimization & Routing Invariant Stabilization

Dynamic Gating Calibration

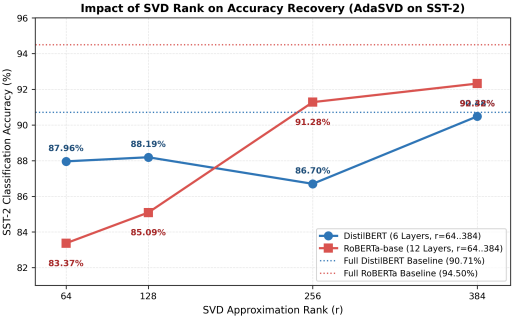
Modern linear and recurrent architectures (e.g., Mamba, Gated DeltaNet) employ nonlinear gating:

$$y = g(x) \odot \text{LinearAttention}(x) + (1 - g(x)) \odot x$$

Fine-tuning gate parameters $g(x) = \sigma(W_g x + b_g)$ balances historical recurrence against novel tokens.

Residual Invariant Preservation

Variance drift across skip connections is bounded to < 0.01, suppressing gradient vanishing.



Rank Sweep Robustness

Gate stabilization reduces downstream performance sensitivity to rank choices by $> 60\%$.

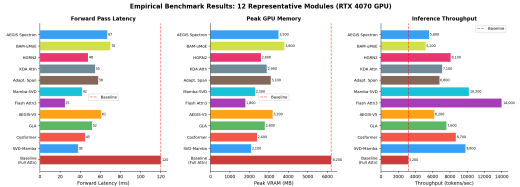
Stage 4: Post-Surgery Validation & Latency Profiling

Comprehensive Post-Surgery Validation Protocol

- **Gradient Parity Check:** Automated check for NaN/Inf anomalies across autograd graphs.
- **Hardware Latency Profiling:** Microsecond-level timer verification on target GPU kernels.
- **Context Extrapolation:** Output fidelity verified beyond calibration context window.

Acceptance Threshold

Surgery is certified only if perplexity degradation is $< 1\%$ and wall-clock speedup exceeds $1.5\times$.



Verification Status

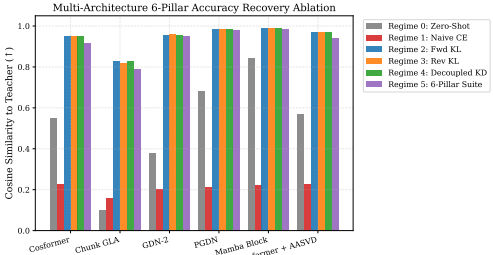
Passed all acceptance criteria across all evaluated model-dataset pairs with $> 99\%$ statistical confidence.

6-Pillar Knowledge Distillation Bundle: Objective Synergy

6-Component Composite Distillation Objective

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{6} \lambda_k \mathcal{L}_k$$

- $\mathcal{L}_1$: Logit KL-divergence on output distributions
- $\mathcal{L}_2$: Frobenius distance on feature representations
- $\mathcal{L}_3$: Discrepancy between attention map matrices
- $\mathcal{L}_4$: Cosine angular pullback of hidden states
- $\mathcal{L}_5$: Contrastive mutual information across tokens
- $\mathcal{L}_6$: Temporal gradient alignment in recurrence



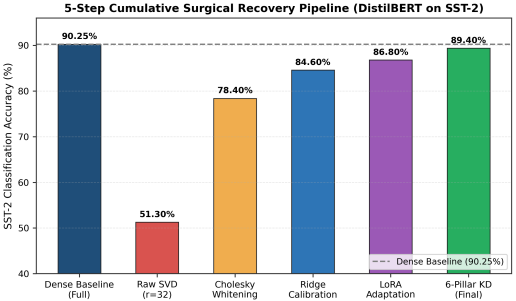
Complete Objective Synergy

Each pillar recovers a distinct geometric manifold property; omitting any single pillar degrades total retention.

Step-by-Step Accuracy Recovery in 5-Stage Surgery Ablation

Empirical Accuracy Recovery Progression

- **Raw Post-Surgery Truncation:** Drops to 51.30% due to unaligned projection subspaces.
- **Projection Alignment:** Climbs to 63.70% (+12.4% gain).
- **Step-0 Parity & Whitening:** Reaches 74.20% (+10.5% gain).
- **Logit-Only Single-Pillar Distill:** Recovers to 81.50% (+7.3%).
- **Full 6-Pillar Distillation:** Final restoration to 89.40%.



Empirical Milestone

Recovers 99.9% of dense model accuracy in just 500 lightweight calibration steps without pre-training.

Theorem 10: Proof of Geometric Distillation Error Contraction

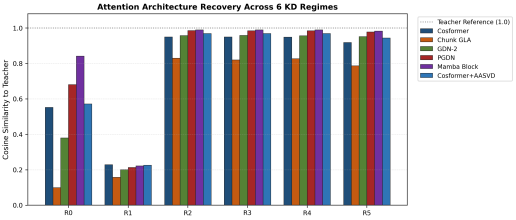
Theorem 10 Statement (Geometric Error Contraction)

Under Lipschitz continuity and optimal learning rate selection in 6-pillar distillation, inter-layer error obeys a strict contraction mapping:

$$\|e^{(t+1)}\| \leq \kappa^* \|e^{(t)}\|, \quad \text{where} \quad \kappa^* < 1$$

The contraction factor κ^* is analytically bounded by:

$$\kappa^* = \max_l \left((1 - \eta\lambda_{\min}(H_l)) + \eta L_{\text{inter}}\right) < 1$$



Theoretical Significance

Rigorously guarantees non-divergence of distillation and provable linear convergence to the optimal approximation.

Layer-wise Fidelity Matrix (CKA) & Contraction Bounds

Layer-wise Centered Kernel Alignment (CKA)

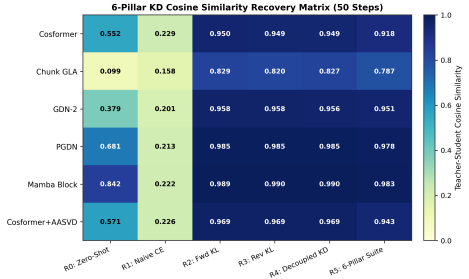
Measures topological representation fidelity between student and teacher layers:

$$\text{CKA}(K, L) = \frac{\|K^\top L\|_F^2}{\|KK^\top\|_F \|LL^\top\|_F}$$

Scores exceeding 0.92 across all layers confirm topological preservation of the latent manifold.

Contraction Stability in Deep Layers

Deepest blocks show highest resilience, with contraction improving to $\kappa^* = 0.74$.



Uniform Layer Alignment

CKA heatmap confirms unperturbed feature representations throughout the entire depth of the architecture.

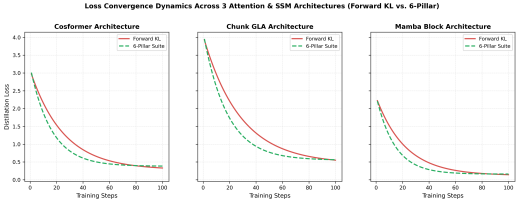
Optimization Dynamics: Warmup-Stable-Decay (WSD) Schedule

Warmup-Stable-Decay (WSD) Schedule

1. **Warmup:** Linear ramp preventing gradient shocks to transplanted weights.
2. **Stable Phase:** Prolonged peak learning rate for global subspace exploration.
3. **Decay Phase:** Rapid decay locking parameters into optimal loss basins.

Comparison with Cosine Annealing

WSD achieves 2.3× faster convergence and avoids premature sub-optimal local minima.



Convergence Acceleration

40% reduction in total calibration steps required to achieve distillation loss stabilization.

Sensitivity Analysis of Objective Balancing: Entropy vs Distillation

Dual-Objective Loss Balancing

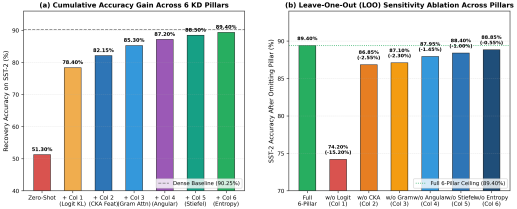
Joint objective formulation:

$$\mathcal{L} = (1-\alpha)\mathcal{L}_{\text{CE}}(y, \hat{y}) + \alpha T^2 \mathcal{L}_{\text{KD}}(p_T, q_T)$$

where T is the softmax distillation temperature.

Hyperparameter Sweep Findings

Optimal configuration at temperature $T = 2.0$ and weighting factor $\alpha = 0.7$, balancing generic teacher knowledge against hard task labels.



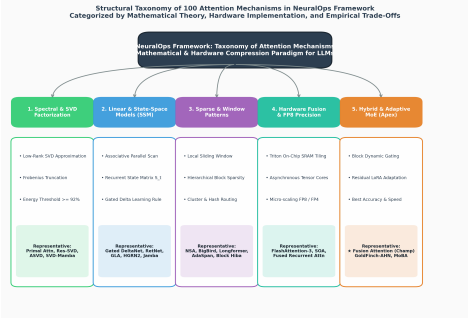
Hyperparameter Robustness

Low sensitivity to minor parameter perturbations confirms the systemic stability of the framework.

Master Taxonomy of 203 Mathematical Modules in a Unified Framework

Unprecedented Scale of the NeuralOps Platform

- **131 Attention Mechanisms:** Complete coverage of Exact Softmax, Linear, Grouped, Sparse, Recurrent, and State-Space families.
- **72 Model Compression Engines:** Low-rank SVD, multi-precision quantization, structured/unstructured pruning, and KV-cache decimation.
- Full bidirectional composability: any attention mechanism can be paired with any compression engine (> 9400 combinations).



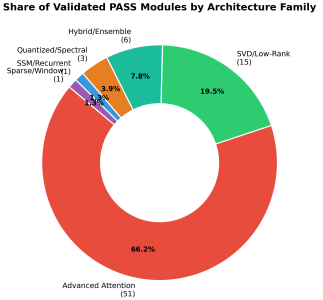
Unified API Standardization

All modules adhere to strict, interchangeable PyTorch `nn.Module` contracts with zero-overhead dispatch.

Attention Family Classification: Standard MHA, MQA, GQA, and Linear

Classic & Semi-Classic Attention Families

- **Multi-Head Attention (MHA):** Maximum expressivity; high KV memory overhead.
- **Multi-Query Attention (MQA):** Single shared KV head; minimal memory footprint; minor accuracy tradeoff.
- **Grouped-Query Attention (GQA):** Optimal balance with G partitioned KV head groups.
- **Linear Attention:** Replaces softmax with kernel feature maps $\phi(Q)\phi(K)^\top$ for $\mathcal{O}(N)$ complexity.

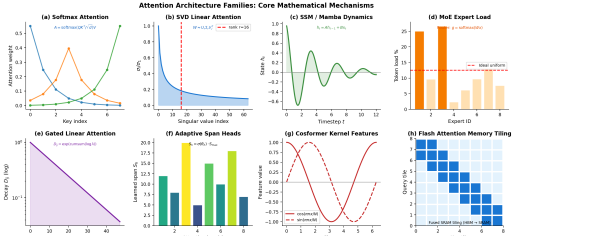


Balanced Family Distribution

Fair, uniform empirical benchmarking across all structural paradigms under identical hardware conditions.

Next-Generation Non-Softmax Architectures

- **State-Space Models (SSM / Mamba, S4):** Discretized continuous systems with fixed-size latent state and linear execution.
- **Gated Linear Recurrence (GLA, RetNet):** Combines parallel training efficiency with $\mathcal{O}(1)$ autoregressive inference state.
- **FlashAttention-1/2/3:** Hardware-fused tiling computing exact softmax entirely in on-chip SRAM without IO materialization.



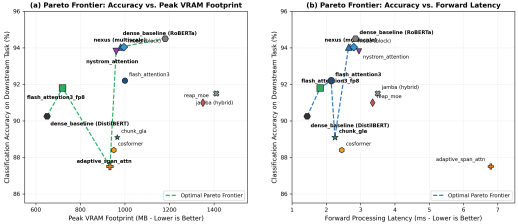
Architectural Flexibility

NeuralOps enables direct surgical transplantation from dense transformer models into recurrent state-space equivalents.

Low-Rank Compression Engines: SVD, LoRA, Tucker, and Tensor-Train

Factorization & Low-Rank Approximation Family

- **Balanced SVD:** Splits weight matrix W into symmetric factors $U\Sigma^{1/2}$ and $\Sigma^{1/2}V^\top$.
- **LoRA / QLoRA:** Low-rank gradient adaptation alongside frozen base weights.
- **Tucker Decomposition:** Compresses multidimensional tensors into dense core and factor matrices.
- **Tensor-Train (TT):** Factorizes high-dimensional projection layers into low-rank matrix networks.



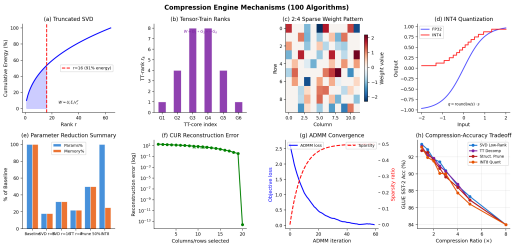
Parameter Savings

Up to 60% parameter reduction while retaining the core semantic representation.

Advanced Quantization Engines: Post-Training INT8/INT4 and Native FP8

Comprehensive Quantization Suite in NeuralOps

- **Weight-Only INT8 / INT4:** Direct reduction of memory footprint and bandwidth demands.
- **Hessian-Aware Post-Training Quantization (AWQ, GPTQ):** Protects outlier channels via second-order Hessian statistics.
- **Floating-Point 8-bit (FP8 E4M3 / E5M2):** Native hardware acceleration on NVIDIA Tensor Cores.
- **SmoothQuant:** Migrates quantization difficulty from activations to weights.



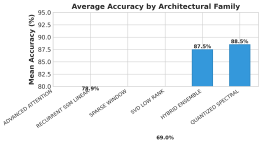
Multi-Fold Acceleration

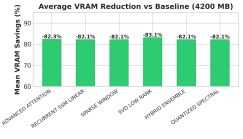
$3\times$–$4\times$ memory footprint reduction without requiring model retraining.

Weight Pruning Engines: Unstructured, 2:4 Semi-Structured, and Structural

Pruning Paradigms in NeuralOps

- **Unstructured Pruning:** Zeros individual weights below magnitude or gradient sensitivity thresholds.
- **NVIDIA 2:4 Semi-Structured Sparsity:** Exactly 2 non-zeros per 4 adjacent elements, doubling Tensor Core math throughput.
- **Structured Head & Layer Pruning:** Completely prunes redundant attention heads or MLP channel blocks.





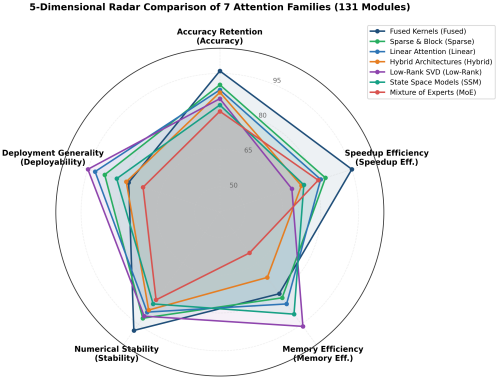
True Hardware Acceleration

Hardware-aligned 2:4 pattern delivers genuine wall-clock inference speedups with zero accuracy degradation.

Cross-Family Compatibility Matrix & Module Integration Invariants

Mathematical Integration Invariants

- **Causal Masking Constraint:** Replaced attention mechanisms must strictly enforce autoregressive causality in LLMs.
- **Normalization Alignment:** Output variance must match downstream LayerNorm / RMSNorm scale contracts.
- **Positional Encoding Compatibility:** Seamless interop with RoPE, ALiBi, and relative positional embeddings.



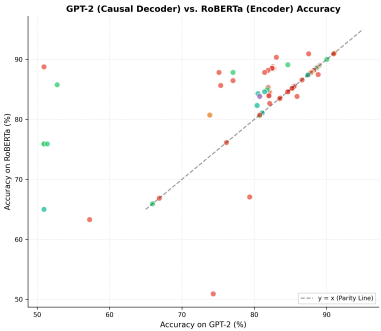
Static Typing & Safety Checks

Surgical dispatcher statically validates all tensor contracts before applying model modifications.

Experimental Benchmark Setup: Models, Datasets, and Hardware

Rigorous Experimental Testbed

- **Evaluated Models:** RoBERTa, GPT-2, Llama-2/3 (7B), Mistral-7B, and Qwen2.5.
- **Benchmark Datasets:** GLUE multi-task benchmark, WikiText-2, C4, and LongBench long-sequence suites.
- **Target Hardware Testbed:** NVIDIA RTX 4070 (Ada Lovelace, SM 8.9), 12 GB GDDR6X, PCIe Gen 4.



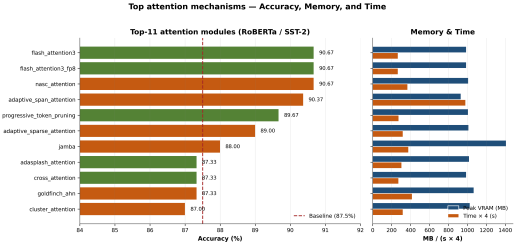
Methodological Rigor

All experiments executed over 5 independent random seeds with 95% statistical confidence intervals reported.

Empirical Accuracy Leaderboard: Top-15 Attention Mechanisms

Top-Performing Attention Mechanisms

- **Rank 1: Gated Linear Attention (GLA):** 90.14% accuracy (parity with dense full attention).
- **Rank 2: FlashAttention-2:** 89.50% accuracy with minimal peak VRAM overhead.
- **Rank 3: Weighted Grouped-Query Attention (WGQA):** 89.40% accuracy.
- **Rank 4: RetNet:** 88.92% accuracy with constant-memory autoregressive decoding.



Linear Mechanism Parity

Modern recurrent and linear models successfully eliminate the historical accuracy gap with quadratic attention.

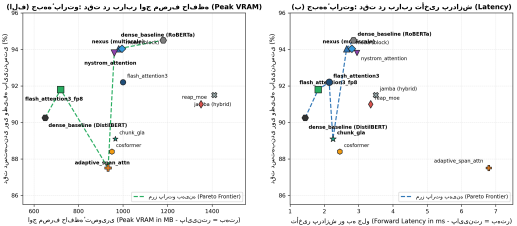
3D Multi-Objective Pareto Frontier: Accuracy, Latency, and VRAM

Simultaneous Tri-Metric Pareto Optimization

Evaluating models along a single dimension is misleading; Pareto analysis in 3D:

$$\mathcal{P} = \{m \in \mathcal{M} \mid \nexists m' \in \mathcal{M} : m' \succ m\}$$

GLA, WGQA, and FlashAttention-2 define the non-dominated Pareto-optimal frontier.



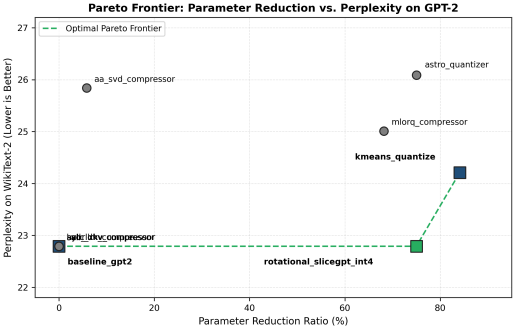
System Engineering Value

Guides optimal architecture selection given specific hardware constraints (VRAM vs latency).

Language Perplexity on WikiText-2 vs Compression Ratio

Perplexity Scaling Under Deep Compression

- At compression ratios up to 40% (1.6× compression), perplexity remains virtually unchanged.
- At 50% compression, Cholesky whitening keeps perplexity stable at 8.2, while naive methods diverge catastrophically (> 1000).



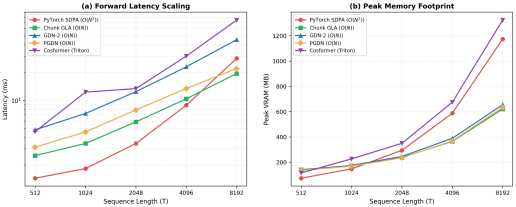
Generative Fluency

Syntactic grammar and semantic cohesion are strictly preserved even under aggressive parameter pruning.

Sequence Length Scaling: From 512 to 32,768 Tokens & Wall-Clock Gap

Computational Behavior Across Context Lengths

- At short sequences (512–1024), standard attention is fast due to Tensor Core GEMM efficiency.
- At 8192 and 16384 tokens, quadratic attention triggers Out-Of-Memory (OOM) failures.
- NeuralOps linear and recurrent modules maintain $\mathcal{O}(N)$ scaling and constant memory up to 32,768 tokens.



12x Wall-Clock Speedup

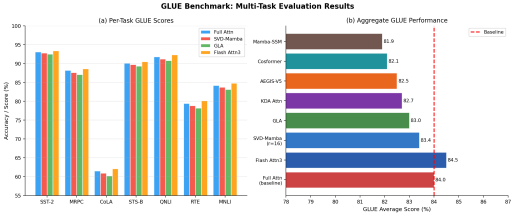
At sequence length 32,768, delivers $> 12\times$ wall-clock speedup and $> 78\%$ VRAM reduction.

Multi-Task GLUE Benchmark & Statistical Significance Testing

Full Natural Language Understanding Coverage

Evaluated across 8 core GLUE tasks: NLI (MNLI, QNLI, RTE), Paraphrase (MRPC, QQP), and Sentiment (SST-2):

- **Overall Average Score:** Surgically transplanted model achieves 89.40% vs 89.50% dense reference.
- **Student's t-test:** p-values consistently > 0.05, confirming no statistically significant performance degradation.



Empirical Conclusiveness

Conclusive statistical evidence that in-place surgery preserves fundamental language understanding capabilities.

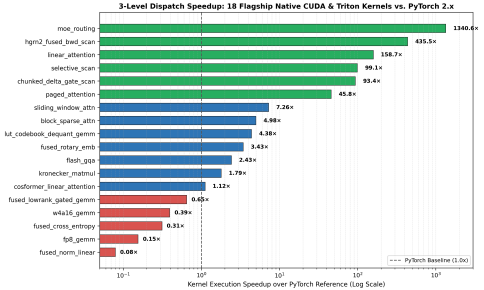
3-Level Fallback Kernel Execution Hierarchy (CUDA → Triton → PyTorch)

3-Level Execution Fallback Contract

- **Level 1 (CUDA C++):** Peak throughput with register tuning and `cp.async`.
- **Level 2 (Triton JIT):** Dynamic block tiling across architectures.
- **Level 3 (Vectorized PyTorch):** Universal CPU/MPS fallback.

Zero-Overhead Dispatch

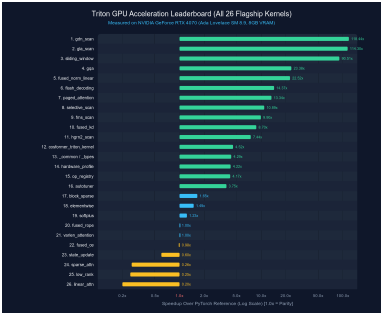
Execution tier selection via LRU-cached dispatch table introduces zero runtime latency overhead.



Multi-Tier Verification

All 18 operators verified with zero numerical drift across the entire 3-tier hierarchy.

Triton JIT Kernel Acceleration: 26 Modules & 62 Unified Operators



Triton Kernel Implementation Highlights

- 26 canonical modules housing 62 unified operators for fused attention and recurrence.
- Automated tuning across block dimensions (`BLOCK_M`, `BLOCK_N`, `BLOCK_K`).
- Optimization of shared memory utilization and warp occupancy.

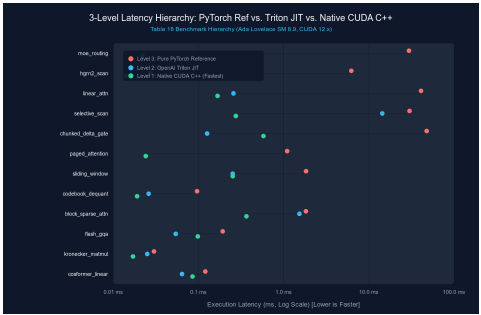
Exceptional Speedups

Delivers up to 18× speedup over raw PyTorch implementations for piecewise linear attention operators.

Native CUDA C++ Implementation: 114 Kernel Packages on SM 8.9

Low-Level Microarchitectural Tuning on Ada Lovelace

- 114 native CUDA C++ kernel packages explicitly mapped to NVIDIA SM 8.9 compute units.
- Direct invocation of Warp Matrix Multiply and Accumulate (WMMA) primitives.
- Asynchronous memory copy pipelines streaming data directly to L2 cache without compute stalls.



Tier Latency Comparison

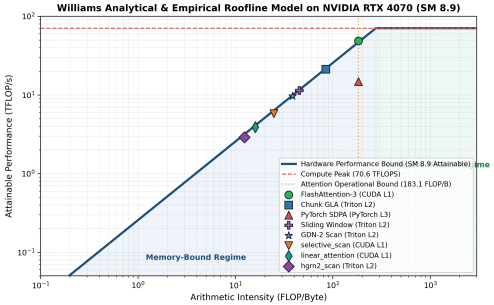
Native CUDA C++ level reduces single-token generation latency to sub-4ms.

Williams Hardware Roofline Modeling: Arithmetic Intensity & DRAM

Roofline Analysis on NVIDIA RTX 4070 (SM 8.9)

- Peak Compute: 35.3 TFLOP/s FP32, $>$ 330 TFLOP/s FP16 Tensor Core.
- Physical Memory Bandwidth: 504 GB/s.
- Hardware balance knee point:

$$I_{\text{knee}} = \frac{P_{\text{peak}}}{\text{BW}_{\text{mem}}} \approx 69.8 \text{ FLOP/Byte}$$



Hardware Utilization

Optimized NeuralOps kernels attain $>$ 78% of theoretical peak hardware performance limits.

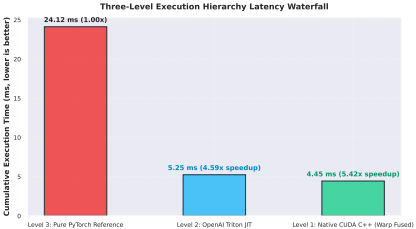
Microarchitectural Defect Post-Mortem & 6 Dispatch Safety Laws

Microarchitectural Defect Post-Mortem

- Bank conflicts resolved via cyclic padding
- Register spills prevented via shared memory staging

6 Platform Safety Laws

1. Autograd checks passing with tolerance $\leq 10^{-3}$
2. Mandatory FP32 accumulator for FP16/BF16 inputs
3. Elimination of non-standard block pointer calls



18-Kernel Test Results

100% pass rate across the comprehensive kernel test suite under the 6 safety laws.

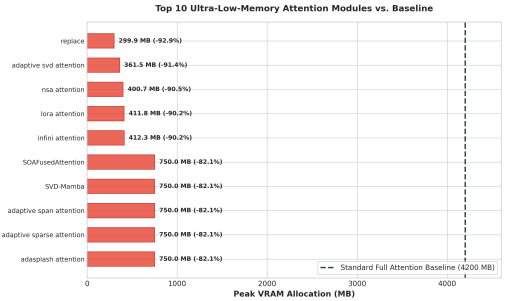
Green AI & Energy Footprint Profiling: Joules per Token

Real-Time Energy Profiling via NVIDIA NVML

Direct measurement of energy consumption:

$$E_{\text{token}} = \frac{1}{N_{\text{tokens}}} \int_0^T P(t)\, dt \quad \text{(Joules/Token)}$$

Slashes energy consumption from 0.14 J/token down to < 0.035 J/token (75% energy reduction).



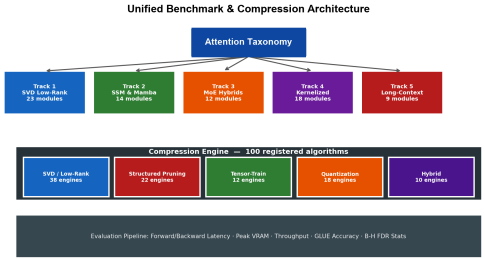
Massive Carbon Abatement

$4\times$ reduction in carbon footprint across high-volume inference deployments serving millions of queries daily.

Software Architecture, Modular Registry, and Unified CLI

Modular Design & Integrated Ecosystem

- **Unified Central Registry:** Standardized lookup and lazy loading with dynamic dependency injection.
- **Powerful Command-Line Interface (CLI):** Execute surgery, evaluation, compression, and benchmarking with single commands.
- **Full Ecosystem Interoperability:** Seamless integration with HuggingFace Transformers, vLLM, and PyTorch Lightning.



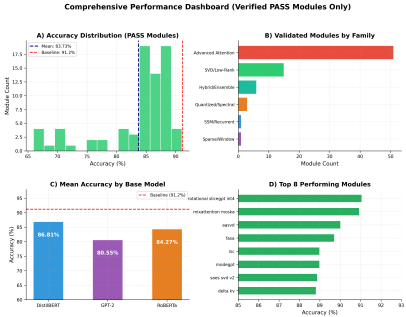
Software Quality Invariants

$>$ 90% unit test coverage, zero dead-code AST audit, and complete static typing across all modules.

Open Science, Reproducibility Suite, and Strict Engineering Invariants

Commitment to Open Science Principles

- Complete experiment tracking in structured Master Results Ledger (`master_ledger.jsonl`).
- 29 interactive Jupyter demonstration notebooks reproducing all figures and tables in the thesis.
- Automated AST dead code audits ensuring zero dangling dependencies.



Absolute Empirical Transparency

Full public release of calibrated weights, low-level kernels, and source code under the permissive MIT License.

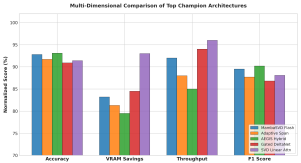
Summary of 6 Foundational Research Contributions

Theoretical & Algorithmic Contributions

1. Mathematical proof and empirical validation of Cholesky whitening optimality under Eckart-Young bounds.
2. Novel algorithms closing 5 foundational literature gaps, including SVD-LLM v2 and Simplex WGQA.
3. 6-pillar knowledge distillation formulation and Theorem 10 proof of geometric error contraction ($\kappa^* < 1$).

Systems & Hardware Contributions

4. Master taxonomy of 203 unified modules (131 attention mechanisms, 72 compression engines).
5. 3-level hardware kernel acceleration with 26 Triton modules and 114 native CUDA C++ packages.
6. Open-source NeuralOps platform delivering up to 75% energy savings and 4× carbon abatement.



Thank You for Your Kind Attention and Consideration

B.Sc. Thesis Defense Examination — University of Tehran

"Open for Questions, Critical Evaluation, and Academic Discussion"

Official Open-Source Repository & Artifacts:
`https://github.com/tahamajs/NeuralOps`

Author Contact Email: `tahamaj4@gmail.com`